

ECON 1550: International Finance

Money, Interest Rates, and Exchange Rates

A model of the money market

Exogenous variables		Endogenous variables			
Variable	Description	Variable	Description	Equation	Type of equation
Y	Real income	R	Domestic interest rate	$M^d/P = L(R, Y)$	Behavioral equation
M^s	Money supply	M^d	Money demand	$M^d = M^s$	Equilibrium condition
P	Price level				

Description of the Money Market

- There are only two assets: money and domestic bonds
- Money
 - Can be used for transactions
 - Pays no interest
- Bonds
 - Cannot be used for transactions
 - Pay interest $i \geq 0$

Money Demand

- Money demand is higher when:
 - Higher price level $P \rightarrow$ need more money to buy goods
 - Lower nominal interest rate $R \rightarrow$ bonds less attractive
 - Higher income $Y \rightarrow$ want more money to buy more
- Capture idea with a behavioral equation

$$M^d = P \times L(\underset{(-)}{R}, \underset{(+)}{Y})$$

Real money demand

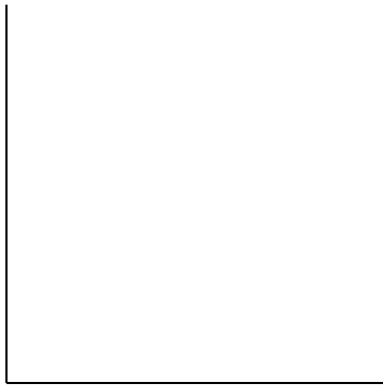
- Convenient to write in terms of real money demand

$$\frac{M^d}{P} = L(R, Y)$$

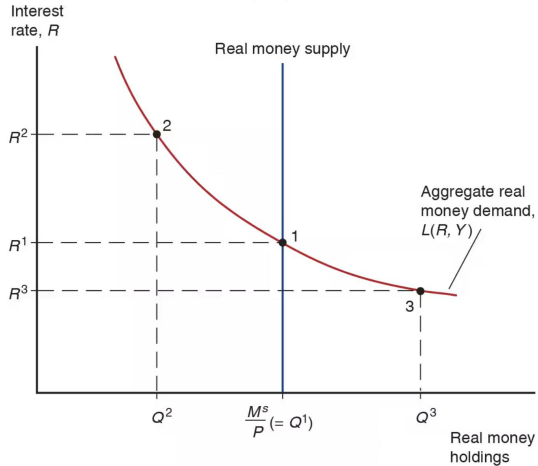
and real money supply

$$\frac{M^s}{P}$$

Equilibrium in money market



Equilibrium in money market



Money is defined by its function

- Medium of exchange
- Store of value
- Unit of account

Many types of money (1/2)

- Commodity money: a physical commodity (like gold) that is used as money
- Convertible paper money: a piece of paper that can be exchanged by a commodity

Many types of money (2/2)

- Fiat money: issued by a central bank, not backed by any commodity
- Digital currency: not backed by any commodity, privately issued, electronic payments

Fiat money

1. Central bank liabilities are special because they are the unit of account.
2. Currency is a promise to deliver future central bank liabilities.

Assets	Liabilities	Assets	Liabilities	Assets	Liabilities
\$0	\$0	\$1 currency	\$1 money	\$1 bonds	\$1 money
$t = 0$		$t = 1$		$t = 2$	

Monetary policy

- Deposits at the Fed are called reserves
- “The” interest rate is just interest on reserves

Short-run FX and money market model

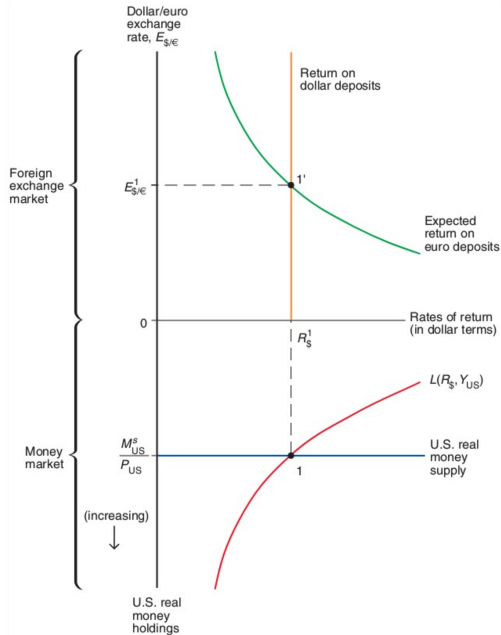
Exogenous variables

Variable	Description
R^*	Foreign interest rate
E^e	Expected exchange rate
Y	Real income
M^s	Money supply
P	Price level

Endogenous variables

Variable	Description	Equation	Type of equation
E	Exchange rate	$R = R^* + \frac{E^e - E}{E}$	Equilibrium condition
R	Domestic interest rate	$M^d/P = L(R, Y)$	Behavioral equation
M^d	Money demand	$M^d = M^s$	Equilibrium condition

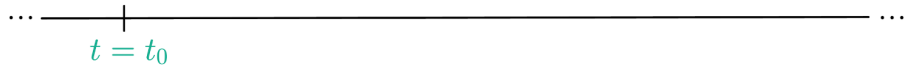
Short-run equilibrium in FX and money markets



Definition of the long-run equilibrium

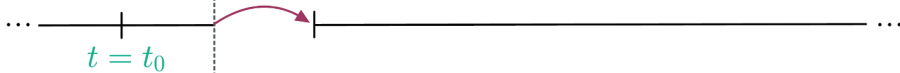
- Hypothetical equilibrium that would result if economy runs indefinitely with no shocks
- Equivalently, the hypothetical equilibrium that would occur if prices were perfectly flexible and always adjusted instantaneously and frictionlessly

Initial long run
equilibrium



Initial long run
equilibrium

Short run



Exogenous
variable
changes

